

Table 1: Reaction rates for the kinetic schemes used in the simulation. [†] SERCA rates reparameterised from the cited source to achieve physiological steady-state ER Ca²⁺ (see Table Notes). Superscripts **Ctrl** / **AD** mark parameters that differ between conditions; all other rates are identical.

Transition / Parameter	Value
Ryanodine receptor^a <i>Saftenku et al. (2001) [1]</i>	
<i>L mode</i>	
$C_{1L} \rightarrow C_{2L}, C_{2L} \rightarrow C_{1L}$	$1.24 \times 10^6 \text{ M}^{-1}\text{s}^{-1}, 13.6 \text{ s}^{-1}$
$C_{2L} \rightarrow C_{3L}, C_{3L} \rightarrow C_{2L}$	$2.98 \times 10^7 \text{ M}^{-1}\text{s}^{-1}, 3867 \text{ s}^{-1}$
$C_{2L} \rightarrow C_{5L}, C_{5L} \rightarrow C_{2L}$	$1.81 \text{ s}^{-1}, 3.63 \text{ s}^{-1}$
$C_{3L} \rightarrow O_{1L}, O_{1L} \rightarrow C_{3L}$	$731.2 \text{ s}^{-1}, 4185 \text{ s}^{-1}$
$C_{3L} \rightarrow O_{2L}, O_{2L} \rightarrow C_{3L}$	$24.5 \text{ s}^{-1}, 156.5 \text{ s}^{-1}$
$C_{3L} \rightarrow O_{3L}, O_{3L} \rightarrow C_{3L}$	$8.5 \text{ s}^{-1}, 111.7 \text{ s}^{-1}$
$O_{2L} \rightarrow C_{4L}, C_{4L} \rightarrow O_{2L}$	$1995 \text{ s}^{-1}, 415.3 \text{ s}^{-1}$
$O_{3L} \rightarrow C_{4L}, C_{4L} \rightarrow O_{3L}$	$253.3 \text{ s}^{-1}, 43.3 \text{ s}^{-1}$
<i>H1 mode</i>	
$C_{1H} \rightarrow C_{2H}, C_{2H} \rightarrow C_{1H}$	$3.26 \times 10^6 \text{ M}^{-1}\text{s}^{-1}, 116 \text{ s}^{-1}$
$C_{2H} \rightarrow C_{3H}, C_{3H} \rightarrow C_{2H}$	$6.6 \times 10^5 \text{ M}^{-1}\text{s}^{-1}, 163 \text{ s}^{-1}$
$C_{2H} \rightarrow O_{1H}, O_{1H} \rightarrow C_{2H}$	$7.86 \times 10^6 \text{ M}^{-1}\text{s}^{-1}, 1480 \text{ s}^{-1}$
$C_{3H} \rightarrow O_{2H}, O_{2H} \rightarrow C_{3H}$	$7.77 \times 10^6 \text{ M}^{-1}\text{s}^{-1}, 330 \text{ s}^{-1}$
$O_{2H} \rightarrow C_{4H}, C_{4H} \rightarrow O_{2H}$	$298 \text{ s}^{-1}, 2390 \text{ s}^{-1}$
<i>L ↔ H1 transitions (same rate for all three pairs)</i>	
$C_{1L} \rightarrow C_{2H}, C_{2H} \rightarrow C_{1L}$	$667 \text{ M}^{-1}\text{s}^{-1}, 0.083 \text{ s}^{-1}$
$C_{2L} \rightarrow C_{3H}, C_{3H} \rightarrow C_{2L}$	$667 \text{ M}^{-1}\text{s}^{-1}, 0.083 \text{ s}^{-1}$
$C_{3L} \rightarrow C_{4H}, C_{4H} \rightarrow C_{3L}$	$667 \text{ M}^{-1}\text{s}^{-1}, 0.083 \text{ s}^{-1}$
Ca ²⁺ flux (k_{Flux} , bidirectional)	$1.09 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$
SERCA pump[†] <i>Higgins et al. (2006) [2]</i>	
$X_0 \rightarrow Y_0, Y_0 \rightarrow X_0$	$4.28 \times 10^{-4} \text{ s}^{-1}, 0.143 \text{ s}^{-1}$
($k_{X1 \rightarrow Y1}, k_{Y1 \rightarrow X1}$)	
$X_2 \rightarrow Y_2, Y_2 \rightarrow X_2$ (occlusion)	$0.43 \text{ s}^{-1}, 15.1 \text{ s}^{-1}$
($k_{X2 \rightarrow Y2}, k_{Y2 \rightarrow X2}$)	
X_{10}, Y_{10} (1 st Ca ²⁺ off)	$77.0 \text{ s}^{-1}, 245.7 \text{ s}^{-1}$
($k_{X1A \rightarrow X1}, k_{Y1A \rightarrow Y1}$)	
X_{12} (Ca ²⁺ on, cytoplasmic; $k_{X1A \rightarrow X2}$)	$8.2 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$
(1 st step: $k_{X1 \rightarrow X1A} = 1.64 \times 10^9 \text{ M}^{-1}\text{s}^{-1}$)	
Y_{12} (Ca ²⁺ on, luminal; $k_{Y1A \rightarrow Y2}$)	$1.5 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$
(1 st step: $k_{Y1 \rightarrow Y1A} = 3.0 \times 10^5 \text{ M}^{-1}\text{s}^{-1}$)	
ER Ca ²⁺ steady-state	$250 \mu\text{M}^{\text{Ctrl}} / 750 \mu\text{M}^{\text{AD}}$
ER Ca²⁺ leak (presenilin)^d <i>Tu et al. (2006) [8]</i>	
Ca ²⁺ leak rate ($k_{\text{ER,leak}}$)	$0.003^{\text{Ctrl}} / 1.80 \times 10^{-4}^{\text{AD}} \text{ M}^{-1}\text{s}^{-1}$
PMCA pump <i>Penheiter et al. (2001) [3]; Brini & Carafoli (2009) [4]</i>	
Association rate (k_{pm1})	$1.5 \times 10^8 \text{ M}^{-1}\text{s}^{-1}$
Dissociation rate (k_{pm2})	20 s^{-1}

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Table 1 (*continued*)

Transition / Parameter	Value
Transition rates ($k_{\text{pm3}}, k_{\text{pm4}}$)	$100 \text{ s}^{-1}, 1.0 \times 10^5 \text{ s}^{-1}$
Leak rate ($k_{\text{pmleak}})^c$	12.5 s^{-1}
VDCC (P/Q-type)	<i>Bischofberger et al. (2002) [5]</i>
$\alpha_{10}, \alpha_{20}, \alpha_{30}, \alpha_{40}$	4.04, 6.70, 4.39, 17.33 ms^{-1}
$\beta_{10}, \beta_{20}, \beta_{30}, \beta_{40}$	2.88, 6.30, 8.16, 1.84 ms^{-1}
V_1, V_2, V_3, V_4	49.14, 42.08, 55.31, 26.55 mV
$K_{\text{Flux}}(V)$	$\frac{A V N_A [B - e^{-V/C}]}{2F [1 - e^{V/C}]}$
	$A = 3.72 \text{ pS}, B = 0.3933, C = 80.36 \text{ mV}$
Calbindin-D28k	<i>Nägerl et al. (2000) [6]</i>
$k_{\text{H}+}, k_{\text{M}+}$	$5.5 \times 10^6 \text{ M}^{-1}\text{s}^{-1}, 4.35 \times 10^7 \text{ M}^{-1}\text{s}^{-1}$
$k_{\text{H}-}, k_{\text{M}-}$	$2.6 \text{ s}^{-1}, 35.8 \text{ s}^{-1}$
Calcium sensor model	<i>Nadkarni et al. (2010) [7]</i>
Association rates ($k_{\text{s}+}, k_{\text{a}+}$)	$6.12 \times 10^7 \text{ M}^{-1}\text{s}^{-1}, 3.82 \times 10^6 \text{ M}^{-1}\text{s}^{-1}$
Dissociation rates ($k_{\text{s}-}, k_{\text{a}-}$)	$2.32 \times 10^3 \text{ s}^{-1}, 13 \text{ s}^{-1}$
a, b	0.025, 0.25
$\gamma, \delta, \varepsilon$	$2 \times 10^3 \text{ s}^{-1}, 4.17 \times 10^{-4} \text{ s}^{-1}, 6.33 \text{ ms}$

Table Notes.

^a **RyR2 channel number.** Control: 150 channels. AD: 450 ($3\times$), consistent with the elevated RyR2 expression reported in Alzheimer's disease; $5\times$ (750 channels) explored as a sensitivity condition. Sub-control sensitivity conditions: $\frac{1}{2}\times$ (75 channels) and $\frac{1}{3}\times$ (50 channels). All kinetic rates and Ca^{2+} flux rates are *identical* across conditions.

[†] **SERCA rates reparameterised.** All 12 kinetic rate constants are identical for Control and AD. They were adjusted from Higgins et al. (2006) so that the pump achieves physiological ER Ca^{2+} concentrations on a 10–15 s timescale. State notation: X = cytoplasmic-facing; Y = luminal-facing; subscript = number of bound Ca^{2+} ions ($X_0 = E_1$ empty, $X_2 = E_1 \cdot 2\text{Ca}$ occluded, $Y_0 = E_2$ empty). Statistical factors of 2 are applied to the first binding step in each direction: $k_{X1 \rightarrow X1A} = 2 k_{X12}$ and $k_{Y1 \rightarrow Y1A} = 2 k_{Y12}$.

^c PMCA leak rate computed dynamically to balance Ca^{2+} influx at rest ($[\text{Ca}^{2+}]_i = 100 \text{ nM}$), yielding $k_{\text{pmleak}} = 12.5 \text{ s}^{-1}$ at PMCA surface density $180 \mu\text{m}^{-2}$ ($K_m = 0.8 \mu\text{M}$). The same value holds for both conditions.

^d **ER Ca^{2+} leak (presenilin).** Presenilin-1/2 (PSEN1/2) functions as a passive ER Ca^{2+} leak channel; familial AD mutations ablate this conductance, leading to ER Ca^{2+} overloading. The Control model uses a leak that equilibrates ER Ca^{2+} at $250 \mu\text{M}$; the AD model uses a greatly reduced (near-zero) leak, raising ER Ca^{2+} to $750 \mu\text{M}$.

References

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